Siddhant Kumar Upmanyu

Senior Platform Engineer · SRE · Cloud Architecture

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SUMMARY

Platform and infrastructure engineer with 4+ years at a smart-home IoT company. Sole engineer responsible for its cloud platform and infrastructure, which supports 10,000+ hubs and about 1 million connected devices. Make the architecture, reliability and cost decisions for the cloud side, and write the protocol and API specs that a core team of about 10 firmware, mobile and backend engineers build against. Took over a manually deployed legacy stack and made it stable, with automated, zero-downtime releases.

SKILLS

Cloud & Infrastructure: Google Cloud Platform (GCP: Compute Engine, Cloud Run, Cloud SQL, Cloud Storage, IAM, VPC), AWS (EKS), Kubernetes (k8s), Helm, Docker, Docker Compose, Terraform, OpenTofu, Infrastructure as Code (IaC), Proxmox, cloud-init

CI/CD & Releases: GitHub Actions, GitOps, blue-green deployments, canary releases, feature flags, automated rollback

Observability: Grafana, Loki, Prometheus, VictoriaMetrics, VictoriaLogs, Grafana Alloy, OpenTelemetry, distributed tracing

Reliability: Incident response, root cause analysis (RCA), high availability, disaster recovery, performance and JVM tuning, cost optimization (FinOps)

Linux & Networking: Linux (Debian, Ubuntu), systemd, Nginx, DNS, TCP/UDP, iptables, nftables, NAT64, rate limiting, DDoS mitigation

Security: TLS certificates, Let's Encrypt, mTLS, PKI, IAM, least privilege, zero trust

Databases & Messaging: PostgreSQL, MongoDB, Redis (Pub/Sub), ClickHouse, database administration, query optimization, indexing, data migration

Languages & Frameworks: Go, Kotlin, Java, Python, Rust, Bash, SQL, C, Spring Boot, gRPC, Protocol Buffers, REST APIs, microservices

Practices: Test-driven development (TDD), Extreme Programming (XP), API and protocol design

EXPERIENCE

eGlu Smart Homes (WiZNSystems)

Bengaluru, India

Senior Systems Architect — **Apr 2026 – Present**
Senior Software Engineer — **Apr 2024 – Mar 2026**
Software Engineer — **Mar 2022 – Mar 2024**

- Write the protocol and API specs that firmware, mobile and backend teams build against before each release. Designed the backend hiring process and hired a backend engineer.
- Made zero-downtime releases the default: blue-green cutovers using iptables, deploy scripts with health checks and automatic rollback, a staging environment that mirrors production, and feature flags for gradual rollouts.
- Fixed the root causes of recurring outages: file-descriptor limits that blocked new threads and SSH logins, a service leaking MongoDB connections, and stale Redis connections that silently dropped messages.
- Took over a manually deployed legacy stack in 2023. Replaced manual uploads over SFTP with automated CI/CD and upgraded Spring 4 on Java 8 to Spring Boot 2.7 on Java 17.
- Set up centralized logging and monitoring with Grafana, Loki and VictoriaMetrics, replacing log files stored on production servers.
- Administer Google Cloud (IAM, networking, storage, compute). Moved the marketing website to its own project, fully managed with Terraform, and reduced compute costs with a 3-year committed use discount.
- Moved device telemetry from raw log files into ClickHouse, cutting storage by over 95% across three datasets (largest: 90 GB to 3.7 GB). Wrote the ingestion service in Go, connected to the main backend over gRPC.
- Moved image data out of MongoDB into Cloud Storage with zero downtime, shrinking the database 54x (3.6 GB to 67 MB) and its memory use by 89% (4.6 GB to 500 MB). Removed redundant indexes and added targeted ones to stop full collection scans.
- Connected IPv6-only cellular hubs to IPv4 backend services with in-kernel NAT64 (Jool and nftables), without adding a proxy. Added kernel-level rate limiting to drop abusive traffic before it reaches the application.
- Built a Go tool for reading production logs over mutual TLS with a private certificate authority, so engineers no longer need SSH access to production.
- Maintain the custom UDP protocol that hubs use to talk to the cloud, and the service behind it, built for peaks of up to 50,000 packets per second. Tuned it for low latency, from kernel packet handling to an adaptive JVM thread pool and allocation-free decoding. Leading R&D on its next-generation replacement.

PROJECTS

hrh — Helm release helper — github.com/sku0x20/hrh

Rust CLI for Helm deployments. Shows a diff before applying a release and rolls back automatically on failure.

avoid — Minimal Linux distribution — github.com/sku0x20/avoid

Bootable Linux distribution based on Void Linux, built for server disaster recovery and small headless machines.

stopgap — Kotlin microservice framework — github.com/sku0x20/stopgap

Built on Helidon SE and Java virtual threads. Published on Maven Central, with unit, integration and Docker end-to-end testing.

PUBLICATIONS

Connection-Agnostic Presence Tracking for Stateless Distributed Backends. Zenodo preprint. doi.org/10.5281/zenodo.21717242

EDUCATION

Bachelor of Computer Applications (BCA) — 2019 – 2022